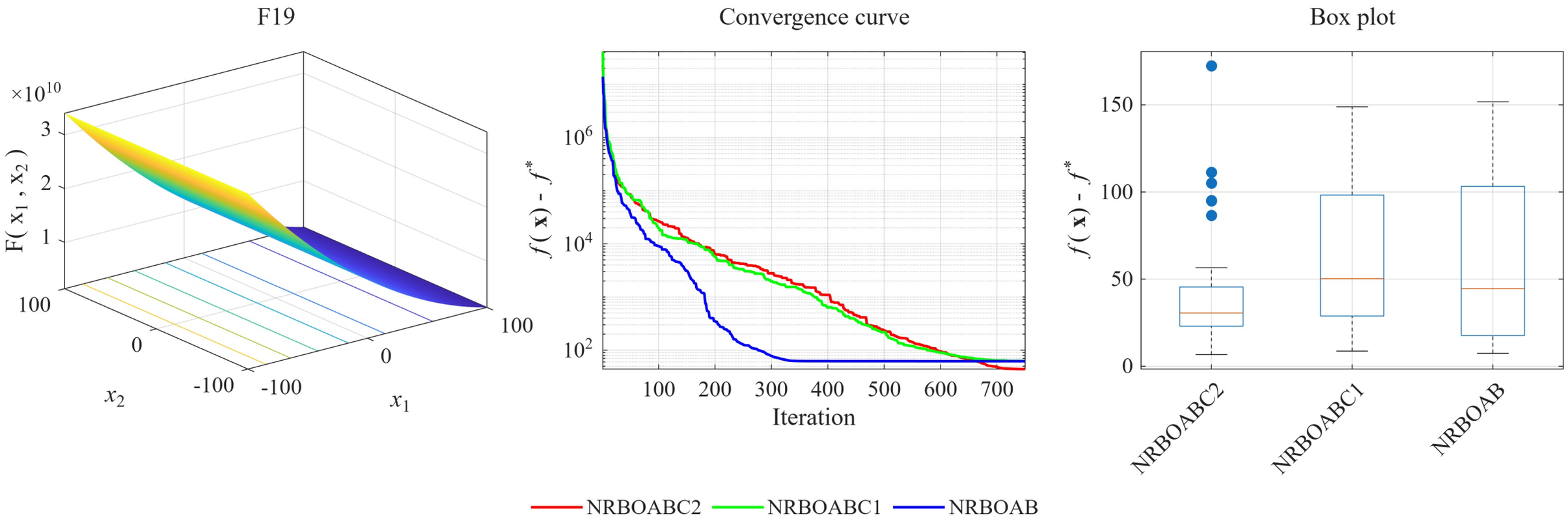

F19
×10^10
3
2
1
F(x1 , x2)
100
0
-100
x2
-100
0
100
x1
Convergence curve
10^6
10^4
10^2
f(x) - f*
100
200
300
400
500
600
700
Iteration
Box plot
150
100
50
0
f(x) - f*
NRBOABC2
NRBOABC1
NRBOAB
NRBOABC2
NRBOABC1
NRBOAB